Anura Development Mode and Chromosome Number Evolution

Blackmon Lab

Evolutionary Biology

Amrutha Kosanaum, Heath Blackmon

Department of Biology

TEXAS A&M UNIVERSITY Biology

Background

Anura is a group of species that encompasses frogs and toads. Typically, when thinking of a frog's lifecycle, many people immediately think of tadpoles that grow into an adult frog. However, not all frogs develop in this way. Frogs and toads can thus further be classified based upon their developmental mode, and can be indirect developers or direct developers. Indirect developers have this distinct tadpole stage that metamorphosizes into an adult morphology, while direct developers closely resemble their adult morphology (just much smaller!) shortly after birth.

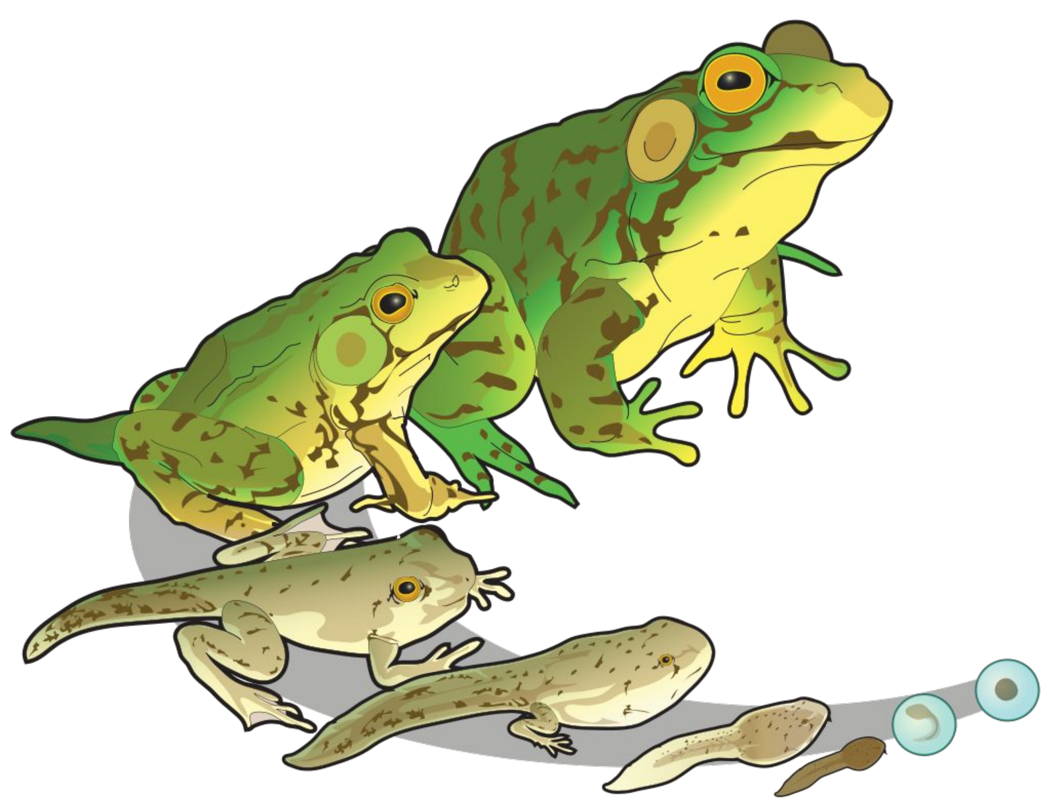

Figure 1. Frog life cycle.
Image credit: Lumen Learning, "The Life Cycle of Amphibians."

Study Design & AI

This project sought to discover how a frog or toad's development mode can affect how many chromosomes they have and how these numbers can change as frogs evolve over time. Coding in the R language was used to run algorithms as well as generate plots to visualize data. Due to the time constraints of a semester-long project, Claude's large language model was utilized to generate R-scripts based on our guided prompts.

First, we used R to map 1,335 Anuran species onto a phylogenetic tree using data from the Blackmon Lab.

Then, we gathered data classifying whether each of these species were indirect or direct developers based on previously available AmphiBIO developmental mode data. We used an AI-generated R-script to optimize our data collection.

Finally, we fit the phylogenetic tree and developmental state data into a Markov Chain Monte Carlo algorithm to analyze the relationships between developmental mode and rates of fission, fusion, polyploidy, and transitions.

Model Fitting

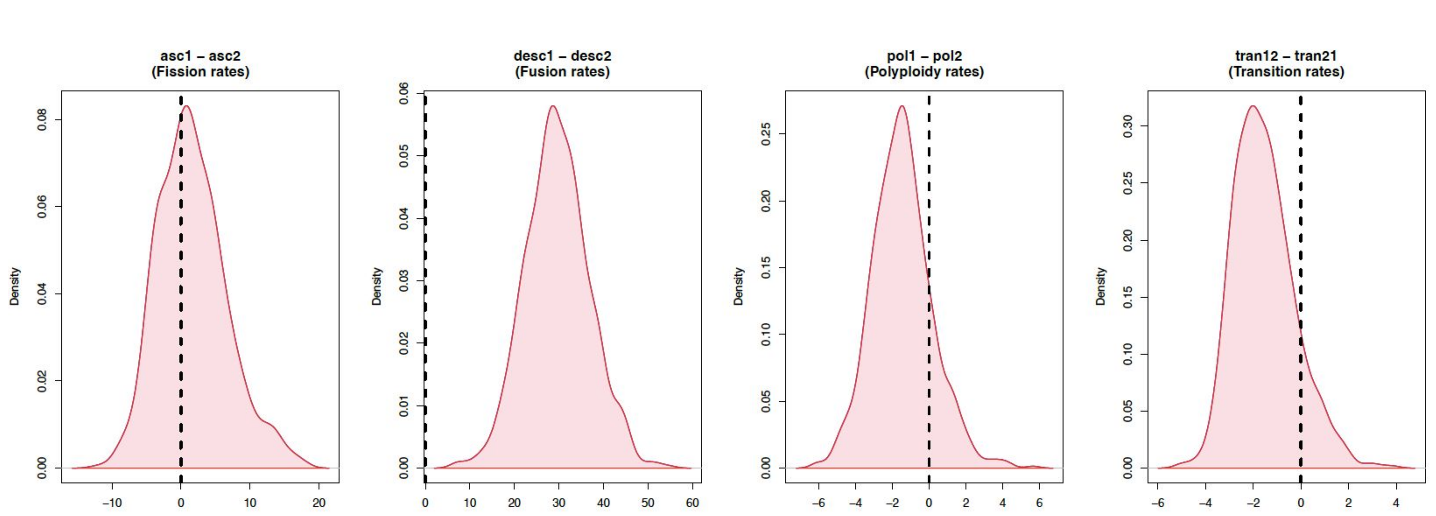

Figure 2. Markov Chain Monte Carlo Plots for Differences in Fission, Fusion, Polyploidy, and Transition Rates.

Results

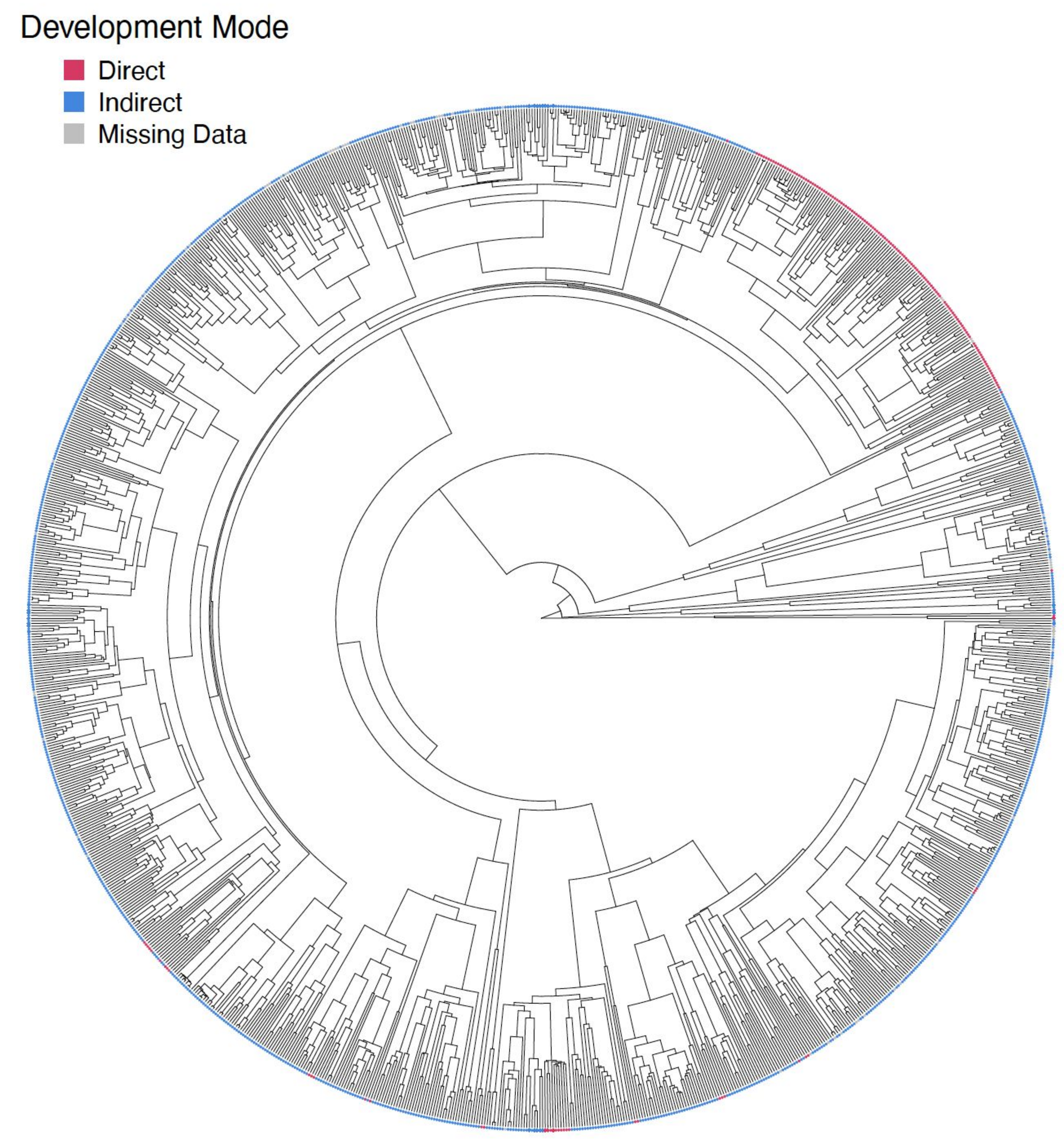

Figure 3. Phylogenetic tree of the Anura clade displaying developmental modes for each taxa.

The plots generated from running our data through the Markov Chain Monte Carlo algorithm demonstrated a significant relationship between fusion rates and indirect developers. On the other hand, the relationships between developmental mode and fission, polyploidy, and transition rates showed little to no significance.

The plot demonstrating the difference in fusion rates elucidates the difference values in fusion rates between indirect and direct developers, which range between 0 to +60. This indicates that the rates of fusion among indirect developers were always higher with the average difference being about 30.

The plots with polyploidy, fission, and transition all center closely around 0, meaning none have a clear relationship with developmental mode.

Discussion

When starting this project, we initially hypothesized that direct developers should have demonstrated higher rates of chromosome number evolution than indirect developers due to smaller effective population sizes. Smaller populations subject direct developers to more genetic drift, which is a change in allele frequencies due to chance events. This idea is called the Drift-Barrier hypothesis.

However, our findings ended up indicating the complete opposite. Indirect developers were shown to have significantly higher rates of chromosome number evolution than direct developers. Specifically, they showed higher rates of fusion. This could be due to a few reasons, though one of the biggest is due to the fact that indirect developers have extremely large populations.

In any population, two competing forces that affect allele frequencies are natural selection and genetic drift. Genetic drift is random and can cause an increase or decrease in beneficial mutations over time, while natural selection is not random, and works to increase the allelic frequency of beneficial mutations over time. Larger populations consist of a larger pool of alleles, allowing more chromosomal mutations to accumulate within the population. Here, there is a higher chance that there are enough alleles with a beneficial mutation that will naturally survive long enough for selection to act on them. This large pool of alleles is also more likely to contain a beneficial allele that is less likely to be lost purely by chance events. On the contrary, smaller populations may develop beneficial mutations as well, but the chances that the individuals with these mutations will survive is much lower due to the stronger influence of genetic drift.

This idea could explain why more rates of fusion are seen in indirect developers. More Anuran species that were indirect developers had a higher chance of having surviving members who could pass on their beneficial mutations to the next generation. Such findings are reinforced by a study done in 2024 by the Blackmon Lab that found a similar pattern with avian chromosome evolution.

In addition, fusion rates evolutionarily tend to be more likely than fission or polyploidy, each for their own reasons. Fission produces a less stable end product because it creates new telomere ends that must be maintained accordingly. Polyploidy is expensive for the cell to perform and can affect reproductive success within a species. Fusion creates a stable enough end product that doesn't always become deleterious for the population.

Conclusions & Acknowledgements

- Indirect developers within the Anura clade were shown to have significantly higher fusion rates than direct developers.
- This could be due to larger population sizes that are more prone to the effects of natural selection on beneficial mutations as opposed to the random effects from genetic drift.

I would like to thank Dr. Blackmon for his guidance, and my classmates for providing feedback throughout this project.